Syllabus for BIOST 561: Computational Skills for Biostatistics

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This syllabus represents the most current information about the course's organization and format. Some of this information may change. I will do my best to be transparent and communicative, and I am grateful for your patience! Please check the announcements page on our Canvas site, and let me know if you have questions via the Canvas Discussion forum.

Last updated April 1, 2026

1 Instructional team & course information

- Instructor
  - Prof. Kevin Lin, Ph.D.
    - Assistant Professor of Biostatistics
    - email: kzlin@uw.edu
    - he/him (Link: what does this mean?)
    - Hans Rosling Center (HRC), Office #338
- Lectures
  - Thursdays 9:30 am - 11:20 am
    - Lectures will be offered in-person in HRC 3rd floor.
    - Everyone's safety is my top priority this quarter. Please do not come to campus if you are sick or have been in contact with someone sick. I am strongly committed to working with you to find ways to help you cover any missed material. Please see the course Illness Policy.
    - Recordings of lectures will be posted to Canvas
- Office hours
  - Wednesday 12:00pm - 1:00pm in Hans Rosling Center (HRC), Office #338

Unless otherwise noted, all times are Pacific Standard Time.

2 Course description and learning objectives

This course treats advanced programming and computing skills for biostatistics. We introduce a sophisticated use of `R` and other programming techniques and tools, including an introduction to using UW Biostatistics' computing resources.

The only way to become a good programmer is through practice. BIOST 561 lectures will focus on concepts and principles, including examples of how to apply them. Stated differently, this class will *guide* your learning, but it will not "teach" you how to program. Most of your learning will happen when you complete your homework. If you get stuck on the homework, resources to help you include the internet (including ChatGPT, StackOverflow, and R and R package documentation!), office hours, and the discussion forum. See "Policy regarding ChatGPT" below for tools regarding AI-based tools.

The following specific topics are covered in BIOST 561:

- Separating product from process and gaining familiarity with RStudio and R projects
- Using Git and GitHub for version control and code distribution and integrating these tools into your typical coding workflow
- Writing R packages, including writing unit tests to ensure your code is properly maintained
- `tidyverse`, including `dplyr`, `magrittr` (i.e., pipes `%>%`), `tibble`, and `ggplot`
- Comfort in writing functions, debugging, profiling code for speed and memory
- Writing simulation studies
- Basic familiarity with using the terminal
- Submitting code to be run on the server (specifically, the Bayes Biostatistics server)
- Creating a Pkgdown website and a personal website via Github.io

3 Course materials

This syllabus, the lecture slides, the exercises, the homework materials, and other resources will be posted on our Canvas site. They will also be posted on the GitHub public repository, https://github.com/linnykos/561_s2025_public.

4 Prerequisites

You must be a PhD or MS candidate in UW Biostatistics to enroll in the class. **You will not be able to pass the class if you are not a Biostatistics student.** This is because you will need access to the UW Biostatistics cluster, which is not possible to obtain without departmental affiliation.

Due to resource constraints, no auditors are permitted.

5 Course grades and homeworks

BIOST 561 is a 2-credit class C/NC. This course cannot be taken for a grade out of 4.0. The grading options are "Credit" or "No Credit".

- Approximately 4 homework assignments will be due over the quarter (every two weeks).
- There will be a final project, where you will create unit tests, documentation, a vignette, and a PkgDown website for any R project you choose (ideally, an R package you are personally invested in).
- To obtain credit for the class, you must make a good faith effort to answer every question on each homework.
- Your homework submissions will be read, commented on, and graded either credit/no credit.
- The homeworks aims to allow you to implement the ideas covered in the lecture. *Most of your learning will happen through working on the homeworks.*
- I may ask you to redo and resubmit your homework if you are not demonstrating that you are meeting the learning objectives for this class.
- I will not be posting homework solutions. Instead, please come to office hours to discuss past homeworks. I am *very* happy to give detailed feedback and suggestions on homeworks during office hours!

6 Protocol for the course

6.1 Lecture protocol

The lectures will be recorded via Panopto and then distributed via Canvas. These recordings are meant to accommodate students who cannot physically be in the classroom during class (for personal emergencies, sickness, disability, etc.). These recording lectures are not meant to replace your learning experience in person. Our class time will sometimes involve in-class group activities, live demos that students can follow, or explanations using the blackboard. The Panopto recording will not be able to capture these aspects accurately. Your learning is most enhanced in person, so I strongly encourage you to come to class in person as often as possible.

6.2 Homework protocol

Homework will typically be due on Monday afternoons. (Please see the homework's due date for any specific homework.) To submit the homework, we will submit two copies of your homework.

- One copy (the `.Rmd` and `.html` files) will be on Canvas (similar to your other courses)
- The second copy will be done via pushing your `.Rmd` and `.html` files to your private GitHub repository that you will make called `UWBiost561`. You will also invite Kevin Lin (username: `linnykos`) to be a collaborator for your private GitHub repository `UWBiost561`. Then, Kevin will provide feedback to your homework (based on a good faith attempt) via GitHub issues. Kevin will also post your completion status on Canvas. (This will happen within the week after the due date.)

The reason for this slightly convoluted grading protocol is to immerse you in the GitHub workflow. One of the primary goals of this course is for you feel comfortable with using GitHub regularly *and* become convinced that it is self-advantageous for you yourself to religiously use GitHub for every coding project you will partake in the future, big or small. Hence, this course is designed top-down with this goal in mind.

7 Learning environment

I take my role as an advocate for your learning in this class seriously. In addition to providing information, assignments, and activities that will support your education, I will do my best to help maintain the classroom as a supportive learning environment that respects diversity.

I will not enforce rules about not using your phone or laptop during class. However, please be mindful of others – please do not cause distractions for your peers in the classroom. At the same time, I request you bring your laptop to class whenever you can, as there will be some protected time in certain classes where I go over the homework with you to ensure that all the students have a strong start on the homework.

Norms for everyone to follow:

- When in class, be focused, engaged, kind, and collaborative
- Be respectful and encouraging of each other
- Support your classmates in being focused and engaged by limiting distracting activities during lectures

Additional norms for the instructional team:

- Make class fun!
- Showcase and teach contemporary coding practices

8 Course communication

8.1 Course updates

I will post any updates for the class using the Canvas Announcements. Please ensure that your Canvas settings for an "Announcement" will alert you "ASAP."

8.2 Contacting me

If you have a concern at any point in the course, I encourage you to communicate it to me. **Please let me know of ways to improve the effectiveness of the course.** Please advise me of any errors in the course notes and homeworks.

Email: My contact info is on page 1 of this syllabus. I will do my best to respond to emails in a timely manner (within 24 hours). Unfortunately, I cannot respond individually to emailed questions about the course material due to the class size. Please post such questions to the Canvas discussion forum or bring them to office hours.

So that I can prioritize your email, **please put "Biost 561:" in the subject title** and contact me using your UW address (unfortunately, emails from personal accounts often go to my spam!).

Canvas messages: Please choose email rather than Canvas messages to contact me.

Canvas discussion forum: If you have a question about the material from lectures, need to clarify a homework question, or have another question related to the course material, please post your question on the Discussion forum on Canvas. Someone else will likely have the same question as you, and sharing your question with the class allows everyone to benefit. You are encouraged to help answer your peers' questions, too.

- Where possible, I would appreciate it if you let me know your feedback by e-mail or Canvas so I can confirm that I have fully addressed it.

Class Concerns: Furthermore, if you have any concerns about the class, please see the instructor about these concerns as soon as possible. If you are not comfortable talking with the instructor or are not satisfied with the response that you receive, you may contact the Department of Biostatistics Associate Director of Academic Affairs (biostgp@uw.edu). If you are still not satisfied with the response that you receive, you may contact the Department of Biostatistics Chair (bchair@uw.edu).

9 Academic integrity

Students at the University of Washington (UW) are expected to maintain the highest standards of academic conduct, professional honesty, and personal integrity.

The UW School of Public Health (SPH) is committed to upholding standards of academic integrity consistent with the academic and professional communities of which it is a part. Plagiarism, cheating, unauthorized use of artificial intelligence (AI) tools, and other misconduct are serious violations of the University of Washington Student Conduct Code (WAC 478-121). We expect you to know and follow the university's policies on cheating and plagiarism, and the SPH Academic Integrity Policy. Any suspected cases of academic misconduct will be handled according to University of Washington regulations. For more information, see the University of Washington Community Standards and Student Conduct.

In this class, the following specific expectations apply to **homeworks**:

- Please first attempt the homework problems on your own.
- If you are stuck on a homework problem, you are strongly encouraged to use online resources to supplement your understanding of the material.
  - That is, please *first* use publicly available resources (e.g., internet sources) to make progress
  - If you are still stuck, you are welcome to discuss your problem with your classmates or the instructional team
- Conversations regarding homework should *never* involve emailing or sharing code.

It is **not** *a violation of academic integrity to work collaboratively with your classmates on your homeworks, but you are likely to learn more if you attempt the homework by yourself (with the help of the internet and R documentation) first.*

This class is intended to set you up for a productive modern statistical career, and practice is critical for solidifying your skills and understanding. While learning how to code can be very frustrating, every minute you spend learning how to debug your code for homework will help you debug your code in your future career!

10 Policy regarding ChatGPT

Students in this course are encouraged to explore using AI-based tools (such as ChatGPT) when completing assignments. All sources, including AI tools, must be appropriately cited. Use of AI without proper citation will be considered academic misconduct and subject to investigation.

Please note that AI results can be biased and inaccurate. You are responsible for ensuring that the information you use (i.e., results, wording, code, etc.) from AI is accurate. Additionally, pay attention to the privacy of your data. Many AI tools will incorporate and use any content you share, so be careful not to unintentionally share copyrighted materials, original work, or personal information.

Learning how to thoughtfully and strategically use AI-based tools may help you develop your skills, refine your work, and prepare you for your future career. If you have any questions about citations or what constitutes academic integrity in this course or at the University of Washington, don't hesitate to contact me to discuss your concerns.

11 Student wellness and mental health

Take care of yourself. Diminished mental health, including significant stress, mood changes, excessive worry, or problems with eating and sleeping, can interfere with optimal academic performance. The source of symptoms might be strictly related to your coursework; if so, please speak with me. However, problems with relationships, family worries, loss, or a personal struggle or crisis can also contribute to decreased academic performance. As a UW student, you can talk with a confidential mental health counselor on the same day. Husky HelpLine is available to support you with same-day access to a Licensed Mental Health Counselor. In the event I suspect you need additional support, I will express my concerns and the reasons for them and remind you of campus resources (e.g., Husky HelpLine, Counseling Center, etc.) that might be helpful to you. It is not my intention to know the details of what might be bothering you, but simply to let you know I am concerned and that help, if needed, is available. Getting help is a smart and courageous thing to do – for yourself and for those who care about you.

12 Illness policy / Extenuating circumstances policy

Your safety is my top priority this quarter. I know that it can be tiring and frustrating to constantly weigh risks and make decisions about your health and safety. Please know I am here to support you as we navigate this public health and learning environment together! Please reach out to me to discuss your concerns about safety in this class.

Please do not come to campus if you are sick or have been exposed to someone who is sick. I am strongly committed to working with you to find ways to help you cover any material missed due to illness. If you become ill, please notify me as soon as possible so we can work together to ensure you achieve the learning goals for this course.

In the cases of absences that result in extended absences from assessment material, I will work with you to find a fair and suitable alternative to the stated grading policy. This may include alternate assignments or altered weighting of the assessment components. Accommodations for students will not benefit or penalize the grades of students in the class who do not require the accommodation.

13 Inclusion & Diversity

Diverse backgrounds, embodiments and experiences are essential to the critical thinking endeavor at the heart of University education. In SPH, we are expected:

- To respect individual differences, which may include, but are not limited to, age, cultural background, disability, ethnicity, family status, gender, immigration status, national origin, race, religion, sex, sexual orientation, socioeconomic status and veteran status.
- To engage respectfully in the discussion of diverse worldviews and ideologies embedded in course readings, presentations and artifacts, including those course materials that are at odds with personal beliefs and values.
- To encourage students with concerns about classroom climate to talk to their instructor, adviser, a member of the departmental or SPH EDI Committee, the Assistant Dean for EDI, or the program's director.

If you make a request to me and you do not feel that my response has been adequate, please contact

- the Department of Biostatistics Graduate Program Director, Ken Rice, biogpd@uw.edu
- the Biostatistics Graduate Program Director, Minh Vo, biostgp@uw.edu
- the chair of the Biostatistics department, Lurdes Inoue, bchair@uw.edu

14 Accommodations for students

14.1 Access and accommodations

Your experience in this class is important to me. It is the policy and practice of the University of Washington to create inclusive and accessible learning environments consistent with federal and state law. If you have already established accommodations with Disability Resources for Students (DRS), please activate your accommodations via myDRS so we can discuss how they will be implemented in this course.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), contact DRS directly to set up an Access Plan. DRS facilitates the interactive process that establishes reasonable accommodations. Contact DRS at disability.uw.edu.

14.2 Religious accommodations

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at Religious Accommodations Policy (https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/). Accommodations must be requested within the first two weeks of this course using the Religious Accommodations Request form (https://registrar.washington.edu/students/religious-accommodations-request/).

14.3 Bias concerns

The Office of the Dean has a student concern policy, a faculty concern policy and standard HR procedures for staff concerns. Our 2018 climate survey states that most people in SPH do not report bias incidents because they do not know where to go. Students are encouraged to report any incidents of bias to someone they feel comfortable with, including instructors, advisers or department staff. They can email dcinfo@uw.edu for immediate follow up. Bias concerns can be anonymously and confidentially reported via the online form found here: https://sph.washington.edu/about/diversity/bias-concerns. Data is collected by the Assistant Dean for EDI and the Director of Program Operations for Student and Academic Services and tracked for resolution and areas are identified for further training.

14.4 Sexual harassment

Sexual harassment is a form of harassment based on the recipient's sex that is characterized by:

- Unwelcome sexual advances, requests for sexual favors, or other verbal or physical conduct of a sexual nature by a person who has authority over the recipient when:
  - Submission to such conduct is an implicit or explicit condition of the individual's employment, academic status, or ability to use University facilities and services, or
  - Submission to or rejection of the conduct affects tangible aspects of the individual's employment, academic status, or use of University facilities.
- Unwelcome and unsolicited language or conduct that creates an intimidating, hostile, or offensive working or learning environment, or has the purpose or effect of unreasonably interfering with an individual's academic or work performance.

If you believe that you are being harassed, or have observed harassment, you can report it to SPH using the bias concerns link. The University also has designated offices to help you: SafeCampus; Office of the Ombud; Title IX Investigation Office; and University Complaint Investigation and Resolution Office.

14.5 Safety

If you feel unsafe or at risk while taking this or any course, please contact SafeCampus at 206-685-7233 anytime. There, you can anonymously discuss safety and well-being concerns for yourself or others. SafeCampus can

provide individualized support, discuss short- and long-term solutions, and connect you with additional resources when requested. For a broader range of resources and assistance, see the Husky Health & Well-Being website.

14.6 Privacy/FERPA statement

This course will have some synchronous learning material that will be recorded, including audio, video, and computer screens. The recordings will only be accessible to students enrolled in the course to review materials. They will not be shared with or accessible to the public.

The University and Zoom have FERPA-compliant agreements to protect the security and privacy of UW Zoom accounts. Students who wish to be recorded should only defer questions once the recording has ended.

If you have any questions or concerns about being recorded during the lecture, please let me know, and I will do my best to answer your questions and address your concerns.

15 Learning resources

UW has several free resources to support you as a student:

- Tutoring
  - Sometimes resources are available through the Dean's Office or Department of Biostatistics to provide additional tutoring for students experiencing challenges with the material. Please email Kevin if you are interested in this resource.
- UW Student Technology Loan Program
  - Loans laptops for free; bookings are required
  - https://stlp.uw.edu/equipment/laptops
- Statistical Consulting Services
  - Faculty and graduate students in the Consulting Program of the Departments of Biostatistics and Statistics offer free statistical advice to University of Washington faculty, staff, and students
  - 50-minute consultations are available by appointment
  - https://www.stat.washington.edu/news-resources/consulting
- Information Technology (UW-IT)
  - For any technology help, please call 206.221.5000 or email help@uw.edu.
  - http://www.washington.edu/itconnect/
- Emergency Aid
  - Emergency Aid assists students who are experiencing unexpected financial hardships that may disrupt their education or prevent them from earning their UW degree. These hardships include emergency medical/dental costs, housing and living expenses, family emergencies, natural disasters, loss of income, and more. Aid may come in a variety of forms, including grants, loans, and/or campus and community resources.
  - https://www.washington.edu/emergencyaid/seattle/
- Counseling Center
  - For free counseling, assessment, referrals, and other services.
  - http://www.washington.edu/counseling/
- Safe Campus
  - Call SafeCampus at 206.685.7233 anytime – no matter where you work or study – to anonymously discuss safety and well-being concerns for yourself or others. SafeCampus's team of caring professionals will provide individualized support while discussing short- and long-term solutions and connecting you with additional resources when requested.
  - https://www.washington.edu/safecampus/
- Diversity Allies
  - Includes contact information for several networks, including the Q Center, Native and Tribal Relations, Southeast Asian Community, Education & Leadership Network, the Women's Center, and more.

  - http://www.washington.edu/diversity/
- Student Parent Resource Center
  - Offering resources for student parents, including childcare assistance
  - https://www.washington.edu/financialaid/types-of-aid/child-care-assistance/
- Veterans Center
  - Offering support for student veterans and their dependents
  - https://www.washington.edu/financialaid/types-of-aid/veterans-benefits/
- Support for undocumented students
  - https://grad.uw.edu/equity-justice/undocumented-students/

If you are looking for something you don't see offered here, please let me know, and I will do my best to help!

16 Acknowledgments

16.1 Land acknowledgment

The University of Washington acknowledges the Coast Salish people of this land, the land which touches the shared waters of all tribes and bands within the Duwamish, Suquamish, Tulalip and Muckleshoot nations.

You can learn more about the history of Seattle and the Duwamish Tribe here:

- https://www.washington.edu/diversity/tribal-relations/

16.2 Instructional Materials Acknowledgement

The course materials for BIOST 561 have benefited from many instructors' input. I am especially grateful to Amy Willis and Eardi Lila, as well as the faculty whom they themselves thank, including Ken Rice and Mauricio Sadinle. All errors are my own. If you see any errors, please let me know by email: kzlin@uw.edu.